RIDE Rhode Island Department of Education

Release of Spring 2026 RICAS Test Items from the Grade 6 Mathematics Paper-Based Test

Spring 2026
Rhode Island Department of Education

RIDE Rhode Island Department of Education

Rhode Island Comprehensive Assessment System
Grade 6 Mathematics Reference Sheet

CONVERSIONS

1 cup = 8 fluid ounces
1 pint = 2 cups
1 quart = 2 pints
1 gallon = 4 quarts
1 gallon ≈ 3.785 liters
1 liter ≈ 0.264 gallon
1 liter = 1000 cubic centimeters

1 inch = 2.54 centimeters
1 meter ≈ 39.37 inches
1 mile = 5280 feet
1 mile = 1760 yards
1 mile ≈ 1.609 kilometers
1 kilometer ≈ 0.62 mile

1 pound = 16 ounces
1 pound ≈ 0.454 kilogram
1 kilogram ≈ 2.2 pounds
1 ton = 2000 pounds

AREA (*A*) FORMULAS

square . . . . . $A = s^2$

rectangle . . . . . $A = bh$

OR

$A = lw$

parallelogram . . . . . $A = bh$

triangle . . . . . $A = \frac{1}{2}bh$

(b = length of base; h = height)

VOLUME (*V*) FORMULAS

right rectangular prism . . . . . $V = lwh$

(l = length; w = width; h = height)

OR

$V = Bh$

(B = area of base; h = height)

Grade 6 Mathematics
SESSION 1

This session contains 10 questions.

You may use your reference sheet during this session.
*You may **not** use a calculator during this session.*



Directions

Read each question carefully and then answer it as well as you can. You must record all answers in this Test & Answer Booklet.

For some questions, you will mark your answers by filling in the circles in your Test & Answer Booklet. Make sure you darken the circles completely. Do not make any marks outside of the circles. If you need to change an answer, be sure to erase your first answer completely.

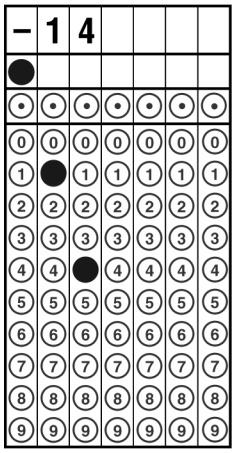
For other questions, you will need to fill in an answer grid. Directions for completing questions with answer grids are provided on the next page.

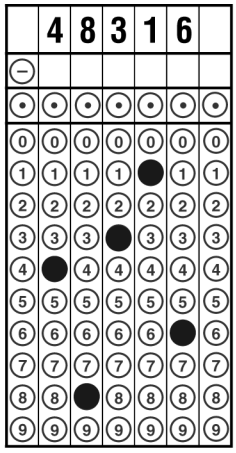
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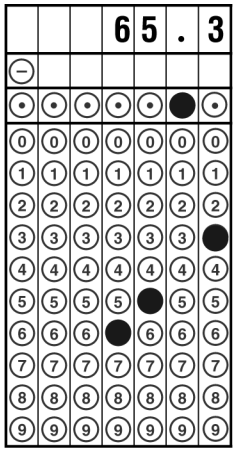
Directions for Completing Questions with Answer Grids

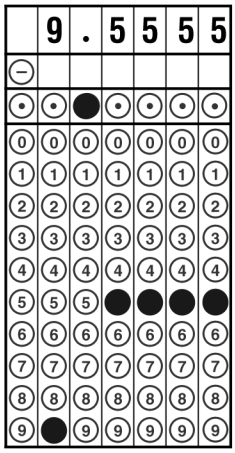
1. Work the question and find an answer.
2. Enter your answer in the answer boxes at the top of the answer grid.
3. Print only one number or symbol in each box. Do not leave a blank box in the middle of an answer.
4. Under each answer box, fill in the circle that matches the number or symbol you wrote above. Make a solid mark that completely fills the circle.
5. Do not fill in a circle under an unused answer box.
6. Fractions cannot be entered into an answer grid and will not be scored. Enter fractions as decimals.
7. If you need to change an answer, be sure to erase your first answer completely.
8. See below for examples of how to correctly complete an answer grid.

Examples









1 A school store sells boxes of colored pencils. Each box of colored pencils contains 24 pencils.

Which of the following expressions represents the total number of pencils in p boxes?

Ⓐ $\frac{24}{p}$

Ⓑ $24p$

Ⓒ $24 - p$

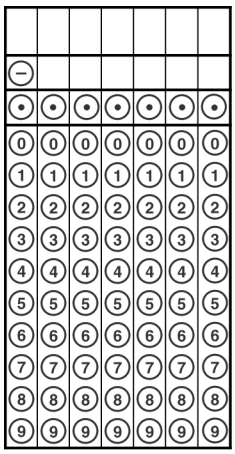
Ⓓ $24 + p$

2 Consider this expression.

$$438.490 - 26.026$$

What is the value of the expression?

Enter your answer in the answer boxes at the top of the answer grid **and** completely fill the matching circles.



3 Clifford makes a necklace using pink beads and orange beads. He uses 4 pink beads and 2 orange beads to make the necklace.

Which of the following statements about the ratios of beads that Clifford uses are true?

Select the **two** correct answers.

Ⓐ The ratio of orange beads to pink beads is 4:2.

Ⓑ The ratio of pink beads to orange beads is 4:8.

Ⓒ The ratio of pink beads to orange beads is 2:1.

Ⓓ The ratio of pink beads to the total number of beads is 4:6.

Ⓔ The ratio of orange beads to the total number of beads is 2:4.

4 Consider this word expression.

A student knits 3 scarves in 4 weeks.

At this rate, how many scarves does the student knit per week?

Ⓐ 0.50

Ⓑ 0.75

Ⓒ 1.0

Ⓓ 1.3

5 A polygon graphed on a coordinate plane has vertices at these points.

(2,–2), (–6,–2), (–6,4), (2,6)

Which of the following **best** describes the polygon?

Ⓐ square

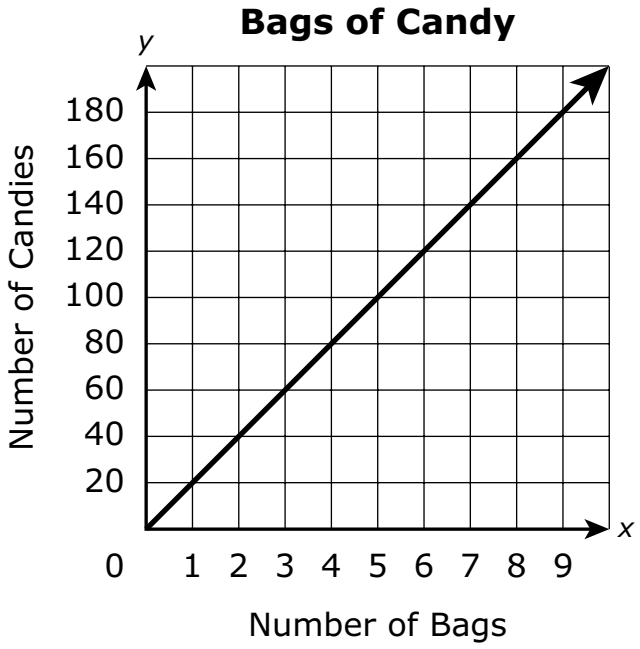
Ⓑ rhombus

Ⓒ trapezoid

Ⓓ rectangle

6 A store owner sells bags of candy. Each bag contains the same number of candies.

This graph shows the relationship between x, the number of bags of candy, and y, the total number of candies contained in the bags.



Based on the graph, which equation can be used to find y, the total number of candies contained in x bags?

Ⓐ $x = 20y$

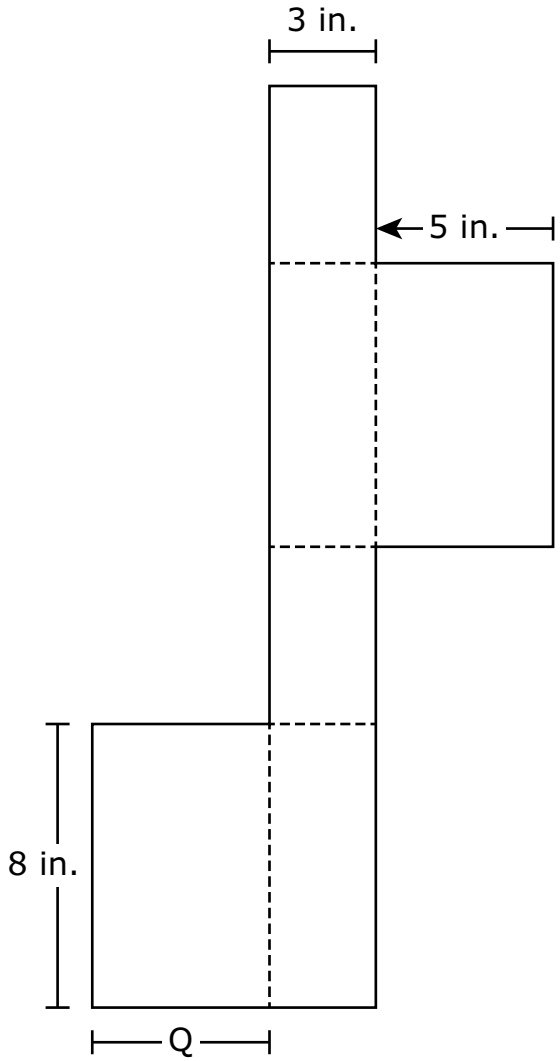
Ⓑ $x = 20 + y$

Ⓒ $y = 20x$

Ⓓ $y = 20 + x$

This question has four parts. Be sure to label each part of your response.

7 A carpenter is making a box. Each side of the box is in the shape of a rectangle.



A. What is the mathematical name of the three-dimensional figure that the net represents?

Mark your answer by filling in the appropriate circle on the next page.

A cube

B square pyramid

C triangular prism

D right rectangular prism

B. Based on the diagram, what is the length, in inches, of side Q?

C. What is the total surface area, in square inches, of the box? Show or explain how you got your answer.

D. The carpenter will cover the entire outer surface of the box with tiles, without gaps or overlaps. The cost of the tiles is $0.05 per square inch.

What is the **total** cost, in dollars, of the tiles needed to cover the entire outer surface of the box? Show or explain how you got your answer.

Write your answers on the next page

7 A. Ⓐ cube

Ⓑ square pyramid

Ⓒ triangular prism

Ⓓ right rectangular prism

8 There are 610 sixth-grade students at a school. This summer, 50 sixth-grade students from the school will volunteer at the public library.

What is the ratio of the number of sixth-grade students who will volunteer at the public library to the total number of sixth-grade students at the school?

Ⓐ 50:560

Ⓑ 50:610

Ⓒ 560:50

Ⓓ 610:50

This question has two parts.

9 Alexander and his two sisters bought a gift for their mother. The 3 children shared the cost equally, and each of them paid more than $12. The total cost of the gift was x dollars.

Part A

Which of the following best represents x, the total cost, in dollars, of the gift?

Ⓐ $x < 13$

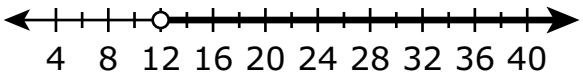
Ⓑ $x > 13$

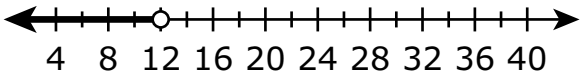
Ⓒ $x < 36$

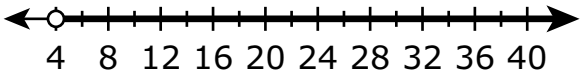
Ⓓ $x > 36$

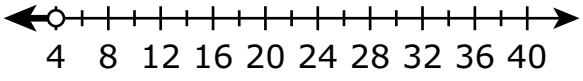
Part B

Which graph represents the amount, in dollars, that Alexander could have paid for his share of the cost of the gift?

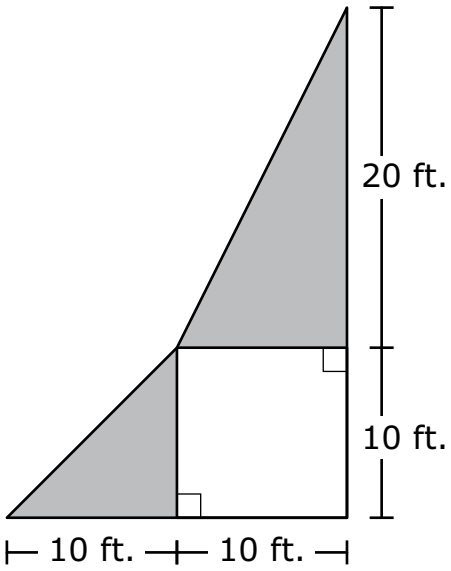
Ⓐ 

Ⓑ 

Ⓒ 

Ⓓ 

10 A figure is decomposed into two triangles and a square. The two triangles are shaded, as shown in this diagram.



Based on the measurements in the diagram, what is the total area, in square feet, of the **triangles**?

Ⓐ 250

Ⓑ 150

Ⓒ 100

Ⓓ 50

Grade 6 Mathematics
SESSION 2

This session contains 10 questions.

You may use your reference sheet during this session.
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Directions

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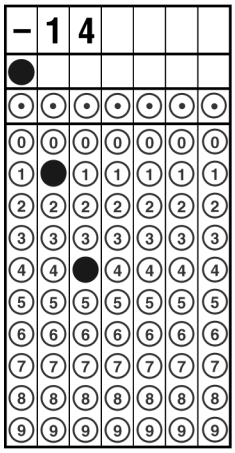
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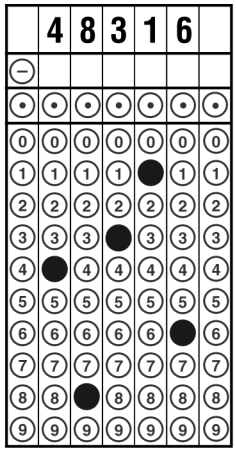
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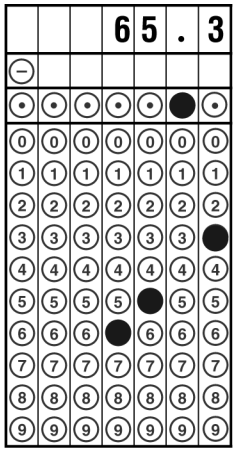
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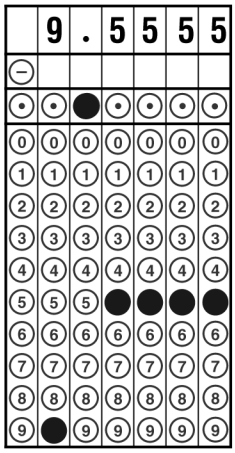
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8. See below for examples of how to correctly complete an answer grid.

Examples









11 A cyclist rode her bicycle a distance of 5 miles.

What is the total distance, in feet, she rode her bicycle?

Ⓐ 1,056 feet

Ⓑ 5,280 feet

Ⓒ 26,400 feet

Ⓓ 52,800 feet

12 Consider this inequality.

$$15 < 3.5y$$

Which of the following values of y make the inequality true?

Select the **two** correct answers.

Ⓐ 2

Ⓑ 3

Ⓒ 4

Ⓓ 5

Ⓔ 6

13 Which of the following is equivalent to 100,000,000?

Ⓐ 10^6

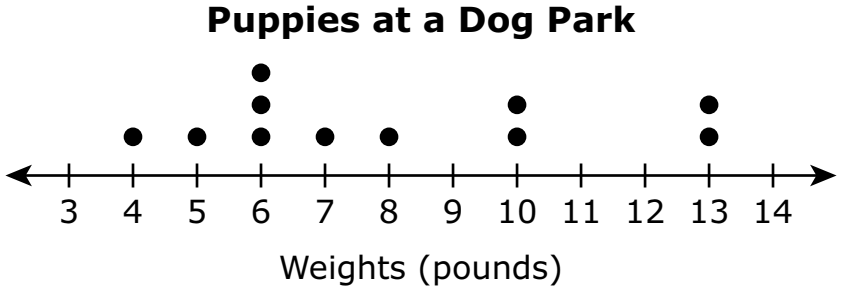
Ⓑ 10^7

Ⓒ 10^8

Ⓓ 10^9

This question has four parts. Be sure to label each part of your response.

14 On Tuesday, a researcher weighed 11 puppies that were playing at a dog park. This dot plot shows the weights, in pounds, of the puppies the researcher weighed.



A. Based on the dot plot, what is the range, in pounds, of the weights of the puppies? Show or explain how you got your answer.

B. Based on the dot plot, what is the median weight, in pounds, of the puppies? Show or explain how you got your answer.

C. Based on the dot plot, what is the mean weight, in pounds, of the puppies? Show or explain how you got your answer.

D. One more puppy arrived at the dog park. The new puppy weighed 8 pounds. The weight of the new puppy will be included in the dot plot.

Which of the measures of the data in the dot plot—the range, the median, or the mean—will change when the weight of the new puppy is included with the other weights? Explain your reasoning.

14

15 An ice skating rink is closed on days when the wind chill temperature is lower than −20°F. At which of the following wind chill temperatures will the ice skating rink be closed?

Ⓐ 24°F

Ⓑ −22°F

Ⓒ −18°F

Ⓓ 0°F

16 Consider this expression.

$$5x + 9$$

Which of the following is equivalent to the expression?

Ⓐ $14x$

Ⓑ $9 + 5x$

Ⓒ $5(x + 9)$

Ⓓ $3(2x + 9)$

17 In 3.5 hours, a carpenter can make 7 birdhouses.

Which of the following statements about the numbers of birdhouses the carpenter can make at this same rate are true?

Select the **two** correct answers.

Ⓐ In 3 hours, the carpenter can make 6 birdhouses.

Ⓑ In 3 hours, the carpenter can make 8 birdhouses.

Ⓒ In 3 hours, the carpenter can make 9.5 birdhouses.

Ⓓ In 5.5 hours, the carpenter can make 6 birdhouses.

Ⓔ In 5.5 hours, the carpenter can make 11 birdhouses.

Ⓕ In 5.5 hours, the carpenter can make 11.5 birdhouses.

18 Consider this expression.

$$20 - 4x$$

Which of the following is equivalent to the expression?

Ⓐ $4(5 - x)$

Ⓑ $4(5 - 4x)$

Ⓒ $5(4 - x)$

Ⓓ $5(4 - 4x)$

19 Each edge length of a cube is $1\frac{1}{2}$ feet. What is the volume, in cubic feet, of the cube?

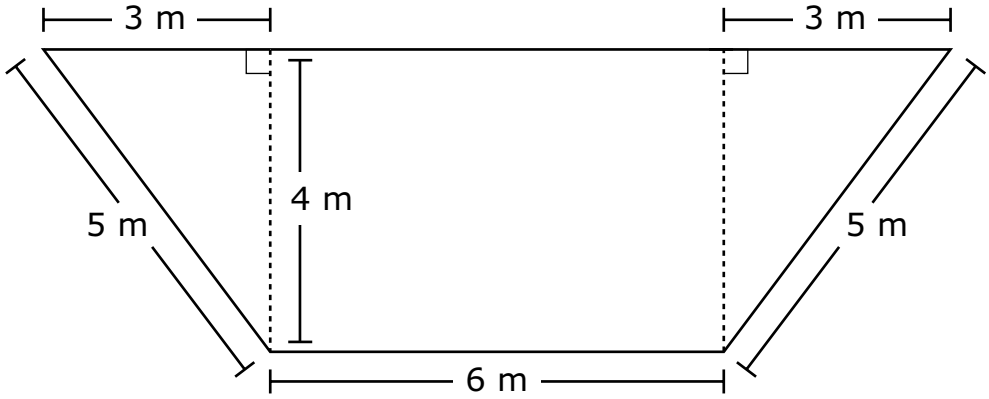
Ⓐ $2\frac{1}{4}$

Ⓑ $3\frac{3}{8}$

Ⓒ $4\frac{1}{2}$

Ⓓ $5\frac{3}{4}$

20 A garden is in the shape of a trapezoid. The garden and some of its dimensions are shown in this diagram.



Based on the diagram, which of the following expressions can be used to find the total area, in square meters, of the garden?

Select the **two** correct answers.

Ⓐ $(6 \times 4) + (3 \times 4)$

Ⓑ $(6 \times 4) + \frac{1}{2}(3 \times 4)$

Ⓒ $(6 \times 4) + \frac{1}{2}(3 \times 5)$

Ⓓ $(6 \times 4) + \frac{1}{2}(3 \times 4) + \frac{1}{2}(3 \times 4)$

Ⓔ $(6 \times 4) + \frac{1}{2}(3 \times 4) + \frac{1}{2}(3 \times 5)$

Grade 6 Mathematics
Spring 2026 Released Operational Items

PBT Item No.	Page No.	Reporting Category	Standard	Item Type*	Item Description	Correct Answer**
1	6	*Expressions and Equations*	6.EE.B.6	SR	Identify an expression with a variable that represents a given real-world context.	B
2	6	*The Number System*	6.NS.B.3	SA	Subtract decimals to thousandths.	412.464
3	7	*Ratios and Proportional Relationships*	6.RP.A.1	SR	Determine whether descriptions of part:part and part:whole ratios are true given a real-world context.	C,D
4	8	*Ratios and Proportional Relationships*	6.RP.A.2	SR	Choose a unit rate that corresponds to a given real-world context.	B
5	8	*Geometry*	6.G.A.3	SR	Determine the type of a polygon given the coordinates of its vertices.	C
6	9	*Expressions and Equations*	6.EE.C.9	SR	Choose a two-variable equation that best represents a given real-world context shown in a graph.	C
7	10–12	*Geometry*	6.G.A.4	CR	Identify a three-dimensional figure from its net and find the surface area of that figure in the context of a real-world problem.	
8	13	*Ratios and Proportional Relationships*	6.RP.A.1	SR	Determine which part:part or part:whole ratio represents a given real-world situation.	B
9	14	*Expressions and Equations*	6.EE.B.8	SR	Identify an inequality that represents a real-world problem and choose a graph that represents the solution set on a number line.	D;A
10	15	*Geometry*	6.G.A.1	SR	Decompose a figure into triangles and squares to solve a mathematical problem involving area.	B
11	18	*Ratios and Proportional Relationships*	6.RP.A.3	SR	Solve a ratio problem using conversion of units within a measurement system.	C
12	18	*Expressions and Equations*	6.EE.B.5	SR	Determine which values of a variable will make an inequality true.	D,E
13	18	*Expressions and Equations*	6.EE.A.1	SR	Determine which expression involving a whole-number exponent is equivalent to a given multi-digit whole number.	C
14	19–20	*Statistics and Probability*	6.SP.B.5	CR	Determine and compare measures of center and variability given a data set and a real-world situation.	
15	21	*The Number System*	6.NS.C.5	SR	Solve a real-world problem involving positive and negative whole numbers.	B
16	21	*Expressions and Equations*	6.EE.A.4	SR	Identify which expression is equivalent to a given variable expression.	B
17	22	*Ratios and Proportional Relationships*	6.RP.A.3	SR	Determine which statements about the unit rate and equivalent ratios in a real-world situation are true.	A,E
18	23	*Expressions and Equations*	6.EE.A.3	SR	Use the distributive property to determine which expression is equivalent to a given variable expression.	A

PBT Item No.	Page No.	Reporting Category	Standard	Item Type*	Item Description	Correct Answer**
19	23	*Geometry*	6.G.A.2	SR	Find the volume of a cube with an edge length expressed as a mixed number.	B
20	24	*Geometry*	6.G.A.1	SR	Determine which expressions can be used to find the total area of a figure.	A,D

* Mathematics item types are: selected-response (SR), short-answer (SA), and constructed-response (CR).

** Answers are provided here for selected-response and short-answer items only. Sample responses and scoring guidelines for any constructed-response items will be posted to the Department's website later this year.